

IMO SDC 12 Meeting Summary

09 February 2026



The 12th session of the IMO Sub-Committee on Ship Design and Construction (SDC 12) was held 19-23 January 2026 at the IMO Headquarters in London. The meeting was supplemented by hybrid (remote) participation. LISCR participated in the following groups in addition to the plenary.

Group	Agenda
WG 1	Working Group on Safe Return to Port, ECRAM and MODU
WG 2	Working Group on FRP, plastic pipe applications and inert gas generation
WG 3	Working Group on GHG Safety
EG 1	Experts Group on URN
DG 1	Drafting Group on ESP Code and other matters

Safe return to the port of passenger ship after a fire or flooding casualty

Interim Explanatory Notes for the Assessment of Passenger Ship Systems' Capabilities After a Fire or Flooding Casualty (MSC.1/Circ.1369)

SDC 12 continued work on the Explanatory Note and developed the draft revised text of the MSC Circular on *Revised Explanatory Notes for the assessment of passenger ship systems' capabilities after a fire or flooding casualty* for approval by 111th session of the Maritime Safety Committee (MSC 111) scheduled for May 2026. The circular is intended to apply to passenger ships for which the building contract is placed on or after 1 January 2028, or, in the absence of a building contract, the keel is laid or the ship is at a similar stage of construction on or after 1 July 2028, or the delivery of which is on or after 1 January 2032. The purpose of the circular is to support harmonised and consistent implementation of SOLAS regulations II-1/8-1.2, II-2/21 and II-2/22 by providing non-mandatory, life-cycle guidance covering the design, assessment, documentation, testing, operation and verification of passenger ship systems following fire or flooding casualties. Structurally, the circular addresses system design principles, casualty thresholds, “remain operational” and restoration criteria, assessment and verification processes, documentation requirements at different project stages, testing and trials, operational considerations under degraded conditions, and life-cycle compliance.

Key issues addressed at SDC 12 were:

- Navigation Systems: clarification of the type of compass required, to avoid ambiguity in the term “working compass” and SDC 12 agreed that a properly adjusted magnetic compass is acceptable;
- External Communication: consideration of how external communication requirements should be expressed, including whether to retain detailed equipment lists or move to a goal-based formulation. SDC 12 agreed to keep the list:

- Ballast system: SDC 12 reaffirmed that the ballast system remains operational after a fire and flooding casualty;
- Oily water treatment system: SDC 12 reaffirmed that the oil water treatment system is not subject to Safe Return port requirements; and
- Training issues (referral to the HTW and III Sub-Committees): SDC 12 agreed to keep the Explanatory Notes at a high level, with training and operational aspects to be considered separately by other Sub-Committees upon completion.

Enhance Survey Programme (ESP) for tankers and bulk carriers

Background - ESP Code

SDC 12 recalled that amendments to the 2011 ESP Code to permit the use of remote inspection techniques (RIT) were developed at SDC 11 (January 2025). These amendments were subsequently approved by MSC 110 (June 2025) for adoption at MSC 111.

Draft RIT Guidelines

SDC 12 developed the *Draft Guidelines on the use of Remote Inspection Techniques (RIT) for ESP Code Surveys* for approval by MSC 111 in conjunction with the adoption of the above-mentioned ESP Code.

The Guidelines support the safe and standardized use of remote inspection techniques (RIT) during surveys conducted under the 2011 ESP Code for bulk carriers and oil tankers. The Guidelines are designed to assist flag State Administrations, surveyors, owners and RIT firms when using RIT as an alternative or supplement to traditional means of access for close-up surveys and thickness measurements. The framework explicitly presumes the presence and active involvement of an attending surveyor, consistent with the ESP Code, with RIT used to support inspection and decision-making rather than replace surveyor judgement. The overarching objective is to ensure that inspections conducted using RIT provide at least the same level of assurance as traditional close-up visual inspection.

The Guidelines are structured into eight main sections, covering the full inspection lifecycle: introduction and scope; general requirements; certification of RIT firms; training requirements; conditions and limitations on RIT use; planning and preparation (including risk assessment and inspection planning); conduct of surveys; and reporting. Core technical requirements emphasize real-time video livestreaming as the primary inspection method, equivalency of inspection assurance, certification of both RIT systems and firms by the Administration, and strict conditions under which RIT may or may not be used (e.g. exclusion where significant corrosion, damage or coating breakdown exists). Detailed provisions are included on risk assessment, pre-inspection verification, confirmatory inspections by traditional means, data capture, data integrity, and documentation of findings.

The Guidelines attach various documents. Annex 1 provides a sample RIT Inspection Plan, setting out the minimum information required for planning and executing an RIT-based survey. Annex 2 contains a RIT Pre-Inspection Verification Card, establishing objective tests for image resolution, detail, contrast and colour to verify system performance on board prior to inspection. Annex 3 includes a sample RIT Firm Inspection Report to standardize the reporting of results, limitations and supporting evidence. Annex 4 sets out technical provisions for the assessment and certification of RIT systems and associated digital technologies, using a goal-based framework with detailed functional requirements addressing fitness for purpose, safety, efficiency, data accuracy and integrity, data retention, and thickness measurement capability, supplemented by Appendix 1, which provides a sample form for an RIT system statement of capability.

Amendments to the 2011 ESP Code (next batch) – Ultrasonic testing

SDC 12 considered the proposal on amending the 2011 ESP Code to allow ultrasonic testing as an alternative to the hose test for hatch cover tightness, to remove the term “oiltight”, and to harmonize fracture detection procedures, noting that the supporting test data were insufficient and that further development of ultrasonic testing guidelines was needed; as a result, SDC 12 did not agree to amend the Code to allow ultrasonic testing for hatch cover tightness at this stage or to delete the term “oiltight”, but agreed to amend paragraph 5.4.2 to allow “other equivalent means” for fracture detection.

Inconsistent implementation of resolution MSC.158(78) and the 2011 ESP Code on the use of portable ladders

The work carried forward to SDC 12 addressed a confirmed inconsistency between the 2011 ESP Code and the Technical Provisions for Means of Access for Inspections (resolution MSC.133(76), as amended), particularly regarding the use of portable ladders to access cargo hold side shell frames in single-side skin bulk carriers. While the ESP Code limits portable means of access to a height of 5 metres for statutory surveys, the Technical Provisions permit access up to the upper bracket throughout the ship’s service life, reflecting their different objectives; however, both instruments are mandatory and intended to support close-up inspection, creating an inconsistency in application. SDC 11 held in 2025 acknowledged this divergence and invited the Committee to consider amendments, and SDC 12 accordingly focused on aligning the Technical Provisions with the survey constraints of the ESP Code, and prepared draft amendments to resolution MSC.158(78) on Technical Provisions for Means of Access for Inspections, were drafted restricting portable ladders to not more than 5m in length to align with the 2011 ESP Code for approval by MSC 111 and subsequent adoption.

Revision of SOLAS steering and propulsion requirements for traditional and new technologies

SDC continued work on traditional and new propulsion and steering systems under SOLAS Chapter II-1 and V, as well as associated mandatory standards for new ships.

Draft text for new SOLAS regulations

SDC 12 considered proposals on new SOLAS regulations II-1/28-1 and II-1/29-1, as well as related amendments to existing SOLAS regulations II-1/3, 28, 29 and 30 and V/25 and 26, with the aim of simplifying and clarifying steering and propulsion requirements for both traditional and non-traditional systems. The proposals sought to remove duplication with existing SOLAS provisions, clarify applicability and definitions, adjust redundancy and failure assumptions for modern propulsion and steering arrangements (including podded systems), and align the draft requirements with the proposed International Standards for Ship Manoeuvrability, while avoiding unnecessary prescriptive detail and potential unintended regulatory burdens.

During the discussion, support was expressed for simplifying and streamlining the draft requirements, particularly where duplication could be avoided, but concerns were raised that key concepts such as the “least favourable steering system” were ambiguous and required clarification, especially for ships with multiple propulsion lines (i.e. more than one independent propulsion unit, such as twin shafts or multiple pods). It was also noted that the proposed redundancy assumptions for podded propulsion systems and the definition of “reduced condition” for non-conventional systems required further work to ensure consistency with existing SOLAS treatment of conventional and single-screw ships, and to avoid inadvertently encouraging unnecessary equipment duplication. Caution was further expressed that some simplifications could unintentionally impose more stringent requirements on traditional steering systems without clear

justification, and it was therefore agreed that further detailed technical consideration should be deferred to the next session.

Draft international standards for ship manoeuvrability (ISSM)

SDC 12 recalled that SDC 11 had agreed to develop a new mandatory instrument to support new SOLAS regulations II-1/28-1 and II-1/29-1, while retaining resolution MSC.137(76) and existing SOLAS regulations II-1/28 and 29 for application to existing ships, using technologically neutral terminology applicable to both traditional and non-traditional propulsion and steering systems. It was further agreed to develop a new MSC resolution on International Standards for Ship Manoeuvrability (ISSM), to be refined through further submissions and trial data in line with the agreed roadmap and with a target completion year of 2028.

SDC 12 considered proposals to revise the draft ISSM. In the discussion, support was expressed for the roadmap and principles proposed for developing mandatory ISSM, including the need to align performance criteria with demonstrated capabilities of the existing fleet and to complete the work by 2028. Proposals addressing non-traditional propulsion and steering systems were supported in principle, while noting that certain tests, such as zig-zag manoeuvres, could pose challenges for such arrangements. At the same time, it was emphasized that the available trial data remained insufficient in scale, diversity and representativeness to underpin comprehensive mandatory standards, with particular concerns regarding data availability, confidentiality, harmonization and coverage of ship types. On this basis, SDC 12 cautioned against premature finalization of the standards and highlighted that further technical work and a credible data infrastructure should precede any move towards mandatory application.

Engine control room alert management (ECRAM)

The proposal on engine room alert management is driven by operational safety and human-element concerns highlighted by serious incidents, notably the Viking Sky blackout, where excessive and poorly prioritised engine room alarms overwhelmed watchkeepers and delayed recovery actions.

ECRAM performance standards

Two distinctly different approaches were proposed for developing ECRAM performance standards, namely incorporating engine room alert management into the existing Bridge Alert Management (BAM) framework by modifying resolution MSC.302(87), or developing a new, stand-alone performance standard specifically for ECRAM. SDC 12 agreed in favour of a stand-alone ECRAM performance standard, on the grounds that engine-room alerting is fundamentally different from bridge alerting in terms of operational concept, human-machine interaction and alarm behaviour, and that simply extending the existing BAM framework would not adequately address alert floods, cognitive workload and modern automation practices.

While SDC 12 agreed to work further through intersessional correspondence group (CG), SDC 12, in general, agreed to use the IMO goal-based standards (GBS) format in principle, with Tier IV (prescriptive rules/regulations) requirements developed where necessary, noting that existing bridge alert management standards were too generic and that some prescriptive elements would therefore be required. SDC 12 agreed on core principles for the standards, including a life-cycle and top-down approach, applicability to all ship types, use of stakeholder experience, and alignment with the alert

definitions and priorities in the 2025 Code on Alerts and Indicators, and further identified key quality attributes such as alerts being actionable, timely, reasonable, trustworthy, non-duplicative, understandable and prioritized.

Use of Fibre Reinforced Plastics (FRP) within ship structures

IMO's discussion on the use of Fibre Reinforced Plastics (FRP) in ship structures reflects the difficulty of reconciling innovation with established safety concepts, as FRP behaves fundamentally differently from steel under fire conditions and lacks a mature, internationally accepted testing framework to demonstrate safety equivalence. While FRP offers design and weight advantages, existing fire test methods under the FTP Code are not well suited to assessing the structural fire resistance of FRP elements, particularly for load-bearing applications, and there is no consensus on appropriate performance criteria that would ensure comparable safety without inadvertently weakening existing standards. At the same time, the scope of the work has been deliberately limited to fire safety under SOLAS chapter II-2, excluding elements contributing to global hull strength and lifecycle issues such as recycling, to avoid expanding the output beyond its mandate. As a result, IMO has taken a cautious, step-by-step approach, focusing on clarifying scope, developing credible and technology-neutral testing methods, and preserving safety equivalence, rather than simply permitting the use of FRP in ship structures without robust technical justification.

Subsequently, SDC 12 prepared the *Draft Revised Interim Guidelines for Use of Fibre Reinforced Plastic (FRP) Elements within Ship Structures: fire safety*, for approval by MSC 111. The revised guidelines revoke existing circular (MSC.1/Circ.1574) and aim to support flag State Administrations in approving the use of FRP elements in ship structures under SOLAS regulation II-2/17, using an alternative design and arrangements approach. The guidelines explain the nature and classification of FRP elements, clarify their scope and limitations (in particular that elements contributing to global hull strength are not fully addressed, and the guidelines are not intended to be used for assessing such elements), and provide a structured fire safety assessment framework aligned with the SOLAS fire safety objectives and functional requirements. The annex systematically examines how FRP elements may affect ignition, fire growth, smoke and toxicity, fire containment, firefighting, structural integrity, and means of escape, and describes how equivalence with prescriptive SOLAS requirements can be demonstrated through engineering analysis and fire testing. Overall, it aims to enable a consistent, performance-based evaluation of FRP fire safety, while recognising uncertainties and the need for further experience.

SDC 12 agreed that matters related to recycling and end-of-life aspects of FRP materials would be more appropriately addressed under instruments managed by the Sub-Committee on Pollution Prevention and Response (PPR) or the Marine Environment Protection Committee (MEPC), rather than within the scope of the Interim FRP Guidelines.

Underwater radiated noise

IMO's work on underwater radiated noise (URN) has developed incrementally, beginning with the approval of the non-mandatory Guidelines for the reduction of underwater noise from commercial shipping in 2014 (MEPC.1/Circ.833), which were subsequently revised and approved at MEPC 80 (2023) as MEPC.1/Circ.906 and further revised at MEPC 82 (2024) as MEPC.1/Circ.906/Rev.1. Recognizing the need for additional technical evidence to assess the effectiveness of the Revised Guidelines, MEPC established an experience-building phase (EBP) supported by the Action Plan, under which SDC

leads the technical work, including data collection, modelling and review of contributions, while MEPC retains responsibility for policy direction. Through the EBP, technical support has been strengthened via expert assistance to the Secretariat, targeted workshops and a correspondence group on URN, and SDC is assessing whether the initially planned three-year EBP provides sufficient technical maturity or whether an extension may be required to ensure that future decisions are based on robust and representative data.

Experience-Building Phase (EBP) framework

SDC 12 reviewed the updated experience-building phase (EBP) framework and identified key lessons learned and remaining gaps hindering effective uptake of the Revised URN Guidelines, including the need for a more holistic approach integrating URN and energy efficiency measures, greater availability and sharing of measurement data, reduced cost and technical barriers to URN measurements, increased attention to sensitive sea areas and regional approaches, and capacity gaps in several countries. Noting that these issues could not be fully addressed within the original timeframe, SDC 12 agreed that a two-year extension of the EBP would be necessary to assess implementation on newbuildings, evaluate incentive schemes, advance integration of energy efficiency and URN considerations, support developing countries, and promote a more comprehensive approach to URN, while stressing that such an extension should not delay delivery of the URN Action Plan. SDC 12 therefore, recommended inviting MEPC 84 to extend the EBP by two years and, given the original target completion year of 2026, to request MEPC 85 to extend the target completion year of the output (work programme) to 2028.

SDC 12 also identified the list of main technical aspects on how to reduce URN from shipping, relevant to future policy considerations of MEPC during the proposed two-year extension of the EBP.

Co-benefit of URN and energy efficiency measures

SDC 12 developed the *Draft Technical guidance on co-optimizing energy efficiency and underwater radiated noise at the design and retrofit stage*, for approval by MEPC 84. The circular provides supplementary technical guidance to support the integration of underwater radiated noise (URN) reduction measures with energy efficiency (EE) considerations during ship design and retrofitting. Its objective is to clarify potential trade-offs between URN mitigation and conventional efficiency optimisation, and to promote co-benefits between quieter ship technologies and energy efficiency performance, building on experience from the URN experience-building phase and IMO expert workshops.

The annex sets out design-stage and operational guidance on how URN and EE can be addressed together, including:

- early integration of URN considerations in ship design and retrofit planning;
- propeller, hull form and propulsion system optimisation to reduce cavitation-related noise while maintaining or improving EE;
- use of emerging technologies such as variable frequency drives, controllable pitch propellers and electrification for combined URN and EE benefits;
- alignment of design optimisation with realistic operating profiles rather than maximum-speed assumptions;
- operational measures, onboard monitoring and modelling methods to assess and manage URN alongside EE performance; and
- encouragement of data sharing, case studies and further submissions under the URN experience-building phase to refine and improve the guidance over time

URN study

SDC 12 developed the draft terms of reference for an IMO-commissioned study on underwater radiated noise (URN) emissions for approval by MEPC 84. The Study is intended to support the URN Action Plan by developing baseline, contemporary and future projections of URN from international shipping aligned, where possible, with the Fifth IMO GHG Study. The annex explains the rationale for aligning URN modelling with existing GHG datasets and scenarios to reduce duplication, while recognising that URN modelling remains outside the scope of the GHG Study itself and would require a separate procurement process. It defines the objectives, expected outputs and technical scope of the study, including frequency coverage, spatial and temporal resolution, baseline years, future scenarios to 2050, and comparison with available measurement data, as well as requirements for model validation, uncertainty assessment and consistency with relevant international acoustic standards.

However, undertaking the study is subject to the availability of funding.

Technical assistance

SDC 12 noted that technical assistance on underwater radiated noise (URN) are being supported through the development of the GloNoise URN Toolkit, which is intended to provide tailored modules for different stakeholders, including shipowners and ship crew, and that Member States seeking assistance on URN may submit formal requests through the Technical Cooperation Request Management System (TCRMS) via the IMO Technical Cooperation Portal. Details can be found on the IMO website: https://glonoise.imo.org/?utm_source=chatgpt.com

Safety regulatory framework to support the reduction of Greenhouse Gas emissions

Following the establishment of the GHG Safety Working Group at MSC 109, MSC 110 endorsed its recommendations on addressing safety barriers and gaps for alternative fuels and new technologies and instructed the SDC Sub-Committee to take forward the tasks assigned to it with flexibility, develop a holistic work plan, begin work on the listed tasks at this session where possible, consider establishing a correspondence group if needed, and report back to MSC 111.

Work plan

SDC 12 focused on shaping a realistic road map and prioritisation rather than advancing regulatory text. While there was general agreement that the Nuclear Code and SOLAS chapter VIII are outdated and will eventually require revision, SDC 12 did not elevate nuclear propulsion as a near-term priority and developed a balanced work plan, taking into account overall workload and the criteria set by MSC 110.

Nuclear power for ships

There was support for moving from the prescriptive Nuclear Code toward a goal-based, technology-neutral framework aligned with MSC guidance, to better accommodate future reactor technologies. A few delegations, however, favoured maintaining the current prescriptive approach due to the specialised and high-risk nature of nuclear propulsion. SDC 12 was unable to progress the issue further due to time limitations.

Future work

SDC 12 agreed to the establishment of a correspondence group to support further work on nuclear power, wind propulsion and wind-assisted power, and lithium-ion battery applications, including collating existing hazard identification information, identifying key technical challenges, reviewing relevant IMO instruments and stability criteria, considering possible amendments to SOLAS regulation II-1/41 for battery use, and reporting back to SDC 13.

Escape Trunk

SDC 12 recalled that earlier discussions at SDC 11 on clarifying the meaning of “lower part” in SOLAS regulations II-2/9 and 13 did not reach consensus, although there was broad agreement that the term refers to the lowest deck level or a platform or passageway, and not simply whichever point is lowest, as given in MSC.1/Circ.1511/Rev.1. While Liberia proposed a new output (work programme) proposing concrete amendments to SOLAS for clarification, MSC 110 instructed the SDC Sub-Committee to consider only any safety concerns associated with the issue.

With regard to the possible amendment to the SOLAS Convention, while many concerns were raised during the discussion, including concerns raised by Liberia, an overwhelming majority, including seafarers’ organizations, expressed no concern over the use of an elevated ladder leading to the entrance to the escape trunk. Subsequently, SDC 12 did not embark on amendments to the SOLAS regulations and recommended that MSC conclude this work.

In this regard, SDC 12 recalled that MSC 110 approved MSC.1/Circ.1689 on an interim basis to clarify escape arrangements and encourage a pragmatic port State control approach until the SOLAS text is amended.

Following SDC 12’s conclusion that no safety issues had been demonstrated and that the output was therefore complete, the Sub-Committee considered the implications for MSC.1/Circ.1689, noting that while the circular remained relevant in encouraging a pragmatic port State control approach in recognition of flag State approvals, its purpose as a temporary measure would cease once the work could be concluded. Accordingly, SDC 12 agreed to invite MSC 111 to revoke MSC.1/Circ.1689 on the basis that the output had been completed.

MODU operation after shutdown

The issue underlying this agenda item arose from concerns that, under Chapter 6 of the 2009 MODU Code, an emergency shutdown scenario may require the de-energization of most electrical equipment on deck due to the possible presence of explosive hydrocarbon fumes, which in practice could render essential equipment unusable. Against this background, MSC 110 decided to develop amendments to clarify the requirements on electrical equipment capable of operation after shutdown and, recognizing that the matter concerns system design, electrical arrangements and construction-level safety considerations rather than operational firefighting or equipment performance alone, effectively shifted the lead responsibility from the IMO Sub-Committee on Ship Systems and Equipment to the SDC Sub-Committee.

SDC 12 prepared draft amendments to chapter 6 of the 2009 MODU Code for approval by MSC, to clarify requirements for electrical equipment intended to remain operable after emergency shutdown, specifying that the term “after shutdown” refers to any stage of an emergency shutdown rather than only total shutdown. The amendments confirm that equipment expected to operate after shutdown should be assessed against each selective shutdown condition, address

ambiguities identified during discussion, and are explicitly limited in application to new MODUs only which is constructed on or after 1 January 2027.

Further information

For further information please contact: imo@liscr.com

Annex

Provisional list of draft resolutions and circulars

- Draft Revised Explanatory Notes for the assessment of passenger ship systems' capabilities after a fire or flooding casualty;
- Draft Guidelines on the use of Remote Inspection Techniques (RIT) for ESP Code Surveys;
- Draft amendments to paragraph 5.4.2 of the ESP Code to allow "other equivalent means" for fracture detection;
- Draft amendments to resolution MSC.158(78) on Technical Provisions for Means of Access for Inspections, to align with the 2011 ESP Code for approval by MSC 111;
- Draft Revised Interim Guidelines for Use of Fibre Reinforced Plastic (FRP) Elements within Ship Structures: Fire Safety; and
- Draft Technical guidance on co-optimizing energy efficiency and underwater radiated noise at the design and retrofit stage.